

ENGINEERING CHEMISTRY LAB			
COURSE CODE	BCHEE102	CIE MARKS	50
TEACHING HOURS/WEEK	0:0:2	SEE MARKS	50
CREDITS	01	EXAM HOURS	3hrs

COURSE OBJECTIVES

CL01	Quantitative analysis of materials by volumetric and chemical methods
CL02	Instrumental methods for developing experimental skills in building technical competence

PART A: Instrumental Experiments

Page

No.

- | | |
|--|-----------|
| 1. Potentiometric estimation of FAS using standard $K_2Cr_2O_7$ solution. | 02 |
| 2. Conductometric estimation of acid mixture. | 07 |
| 3. Determination of Viscosity co-efficient of the given liquid using Ostwald's viscometer. | 12 |
| 4. Colorimetric estimation of Copper | 17 |
| 5. Determination of pKa value of the given weak acid using pH meter | 22 |

PART B: Volumetric Experiments

- | | |
|--|-----------|
| 1. Estimation of Total hardness of water by EDTA complexometric method. | 30 |
| 2. Determination of nickel using EDTA by complexometric method | 35 |
| 3. Determination of percentage of Copper in brass using standard sodium thiosulphate solution. | 39 |
| 4. Determination of COD of industrial waste water. | 43 |
| 5. Estimation of percentage of Iron in the given rust solution using standard $K_2Cr_2O_7$ solution (external indicator method). | 47 |

PART C: Demonstration Experiments

51

1. Flame photometric estimation of sodium and potassium.
2. Synthesis of nanomaterials by precipitation method.

PART - A

EXPERIMENT: 1

DATE:

Potentiometric Estimation of Ferrous Ammonium Sulphate using Standard Potassium Dichromate Solution

Theoretical Discussion:

The variation of potential of an electrode with the concentration of ions with which it is in equilibrium may be used as an indicator in volumetric analysis. The method is applicable for a wide range of titrations, provided an appropriate electrode (indicator electrode) is available. An indicator electrode is the one whose potential indicates the change in concentrations of the ions to be titrated. As it is not possible to determine the electrode potential separately, the indicator electrode is used in conjunction with a reference electrode, the potential of which remains constant during the course of titration. The most commonly used reference electrode is a saturated Calomel Electrode. Suppose, a solution of an acid is titrated with a solution of an alkali, the following set up in the acid solution.

H₂ (Pt) / acid solution // aq. KCl // Calomel electrode

The e.m.f. of such cell is given by

$$E = E_{\text{cal}} - E_{\text{H}} = E + 0.0592 \text{ p}^{\text{H}} \quad \text{at } 25^{\circ}\text{C}$$

The change of p^{H} will be reflected in the change of E . When a small amount of standard alkali is added to the acid, a little change in e.m.f. is produced in the beginning, for the change in electrode potential depends upon the fraction of hydrogen ions removed.

As the equivalence point reaches, the fraction of the hydrogen ions removed by a constant volume of standard alkali increases rapidly, thereby causing a rapid change in e.m.f. Above the equivalence point, there is again a small change in the e.m.f. by the addition of excess alkali. Thus if E.m.f. of the cell is plotted against the volume of the standard alkali added. The point of inflection in the curve (the point where the curve changes its curvature) gives the equivalence point. It may be noted that the change in e.m.f. is much more rapid near the equivalence point than in any other region of the titration before and after the equivalence point.

When the titration curve does not show a sharp inflection point, the exact location of the point of inflection goes rather difficult. The precise method is the differential method where $\Delta E / \Delta V$ values, i.e. change in E (or p^H) resulting from the successive additions of the reagent are plotted against the volume of the reagent added. The maximum of the curve (differential curve) so obtained corresponds to the equivalence point of the titration.

Principle

Potentiometric titration can easily be carried out for any reversible Redox reaction. Consider a platinum electrode dipped in solution of ferrous salt, it assumes a potential which will depend on $\log (C_{Fe^{3+}}/C_{Fe^{2+}})$.



Initially, Fe^{3+} is present in infinitesimal amount which increases when the solution is titrated with an oxidizing agent. This results in a rapid change in the ratio $(C_{Fe^{3+}}/C_{Fe^{2+}})$. Thus, there will be a step at the equivalence point in the E-V curve. Beyond the end point, the e.m.f. depends upon the behavior of the oxidizing agent. In order to make the 'step' at the equivalence point very large and steep, the oxidizing or reducing agents must have widely different redox potentials.

Procedure:

Pipette out 25 cm³ of ferrous ammonium sulphate solution into 250 cm³ beaker. Add dilute sulphuric acid and ion-exchange water to it. Wash and place the **platinum and calomel electrode assembly** and connect to the potentiometer. Read the potential difference between electrodes with the aid of the meter and add 0.5 cm³ of potassium dichromate solution from the burette each time and record the corresponding EMF.

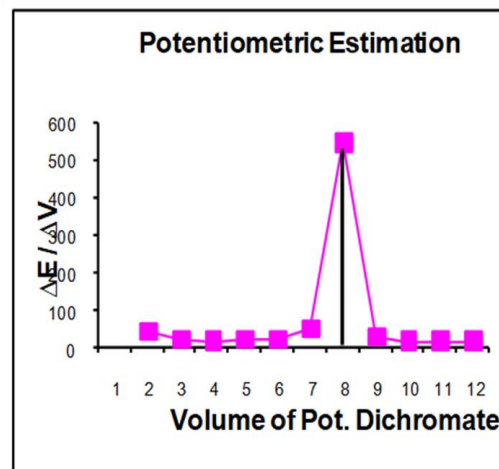
Report: The amount of FAS in the given solution = g

OBSERVATIONS AND CALCULATIONS

Volume of Ferrous Ammonium Sulphate solution = cm³

Normality of K₂Cr₂O₇ solution = N

Volume of K ₂ Cr ₂ O ₇ (V)	EMF (E)	ΔV	ΔE	ΔE/ΔV
0.0		---	---	---
0.5		0.5		
1.0		0.5		
1.5		0.5		
2.0		0.5		
2.5		0.5		
3.0		0.5		
3.5		0.5		
4.0		0.5		
4.5		0.5		
5.0		0.5		
5.5		0.5		
6.0		0.5		
6.5		0.5		
7.0		0.5		
7.5		0.5		
8.0		0.5		
8.5		0.5		
9.0		0.5		
9.5		0.5		
10.0		0.5		



$$\text{Normality of FAS} = \frac{N_{K_2Cr_2O_7} \times V_{K_2Cr_2O_7}}{\text{Vol. of FAS}}$$

$$\text{Weight of FAS / dm}^3 = [N_{FAS} \times \text{Equivalent weight of FAS}]$$

=

= g

Answer the following viva questions

1. Explain the principle involved in potentiometric titration.
2. Name the reference electrode & indicator electrode used in the determination of FAS potentiometrically.
3. Explain the variation of the potential during the titration of FAS > < $\text{K}_2\text{Cr}_2\text{O}_7$ potentiometrically
4. What is the function of calomel electrode?
5. Explain the graph obtained when $\Delta E/\Delta V$ is plotted against volume of $\text{K}_2\text{Cr}_2\text{O}_7$ during the potentiometric titration of FAS > < $\text{K}_2\text{Cr}_2\text{O}_7$.

Conductometric Estimation of Acid Mixture using Standard Sodium Hydroxide Solution

Theoretical Discussion:

The determination of equivalence point of a titration by this method is based upon the measurement of the conductance during the course of titration, which varies in different manner before and after the equivalence point. This is due to the reason that electrical conductance of a solution depends upon the number of ions present and their ionic mobilities. Thus, on plotting the conductance against the volume of the titrant added we get two branches of straight lines; the point of intersection of the lines gives the equivalence point.

Suppose a solution of an electrolyte AB is titrated against a solution of another electrolyte CD, and the following reaction takes place.



If during the titration there is no appreciable volume change and one of the product, say CB, is either slightly ionized or insoluble, then addition of CD will result in a decrease or increase of the conductance of the solution, depending upon the relative mobility of B^- and D^- (B^- is being replaced by D^-). After the equivalence point the conductance changes in different manner depending upon the mobility of C^+ .

Principle:

Let us consider the titration of Acid mixture with NaOH.

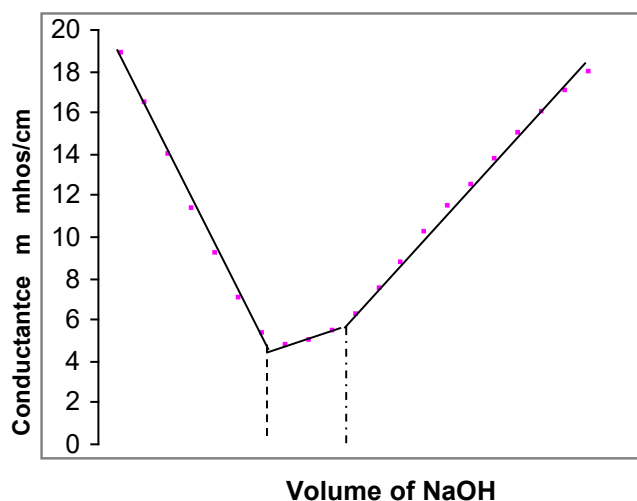
When a mixture containing acetic acid and hydrochloric acid, is titrated against an alkali, strong acid (HCl) will be neutralized first. The neutralization of the weak acid (CH_3COOH) commences after the complete neutralization of strong acid. Thus, the conductance titration curve will be marked by two breaks: the first one corresponds to the equivalence point of HCl and the second to that of CH_3COOH . Let V_1 and V_2 be the volume of alkali corresponding to first and second breaks respectively. Then

$V_1 \text{ cm}^3$ of NaOH required for the neutralization of HCl

$(V_2 - V_1) \text{ cm}^3$ of NaOH required for the neutralization of CH_3COOH

Procedure:

Pipette out 25cm³ of given acid mixture solution into 100 ml beaker. Fill the burette with standard sodium hydroxide solution. Place the conductivity cell inside the beaker. Measure the conductance. Add 0.5 cm³ sodium hydroxide from the burette to the acid mixture solution and record the corresponding conductance.

Nature of Graph:**OBSERVATION AND CALCULATION****Normality of NaOH solution:****N**

Volume of NaOH (cm ³)	Conductivity (ms / cm)	Volume of NaOH (cm ³)	Conductivity (ms / cm)
0.0		5.5	
0.5		6.0	
1.0		6.5	
1.5		7.0	
2.0		7.5	
2.5		8.0	
3.0		8.5	
3.5		9.0	
4.0		9.5	
4.5		10.0	
5.0			

$$\text{Normality of HCl} = \frac{N_{NaOH} \times V_{NaOH}}{V_{HCl}} = \quad \text{N}$$

$$\text{Weight of HCl} / \text{dm}^3 = (N_{HCl} \times \text{Equivalent weight of Hydrochloric acid}) = \quad \text{g}$$

$$\text{Normality of } CH_3COOH = \frac{N_{NaOH} \times V_{NaOH}}{V_{CH_3COOH}} = \quad \text{N}$$

$$\text{Weight of } CH_3COOH / \text{dm}^3 = (N_{CH_3COOH} \times \text{Equivalent weight of Acetic acid}) = \quad \text{g}$$

Report: The amount of HCl in the given solution = g

The amount of CH₃COOH in the given solution = g

Answer the following viva questions

1. Define conductance. Mention its unit.
2. Why the conductance of an acid mixture decreases slightly at first on adding NaOH?
3. Why there is a decrease and increase in conductance is observed
4. State the principle of conductometric titration.
5. Explain the nature of graph

Determination of Viscosity Coefficient of a given liquid using Ostwald's Viscometer

Principle:

Viscosity arises due to internal friction between moving layers of molecules. A liquid flowing through a cylindrical tube of uniform diameter is expected to move in the form of molecular layers. A layer close to the surface is almost stationary while that at the axis of the tube moves faster than any other intermediate layer. A slow moving layer exerts a drag or friction on its nearest moving layer backwards. This property of a liquid by which it retards or opposes motion between layers is called viscosity. The coefficient of viscosity of a liquid is defined as the tangential force per unit area required to maintain a unit velocity gradient between any two successive layers of liquid situated unit distance apart.

The coefficient of viscosity is given by Poiseuille's formula $V = \frac{\pi \rho r^4 \tau}{8 \eta l}$

Where V = volume of the liquid, r = radius of the tube, l = length of the tube, ρ is the pressure difference between the two ends of the tube, η = coefficient of viscosity of the liquid. If equal volumes of two different liquids are allowed to flow through the same tube under identical conditions, then,

$$\frac{n_1}{n_2} = \frac{t_l d_l}{t_w d_w}$$

The time, (t_l) taken by the given liquid to travel through a certain distance in the tube is determined. The time, (t_w) taken by a standard liquid to travel through the same distance is measured. Knowing the densities, (d_l) and (d_w) of the test liquid and the standard and also the coefficient of viscosity of the standard (η_w), coefficient of viscosity of the test liquid (η_l) can be calculated.

Factors affecting viscosity of a liquid:

1. Increase in molecular weight results in an increase in viscosity
2. Branched chain compounds have higher viscosity than those involving straight chains.
3. The polar compounds are more viscous than the non-polar ones. Presence of directed bonds such as hydrogen bond causes the viscosity to increase.

4. The viscosity of liquids increases by the presence of solutes, lyophilic colloids and suspended impurities.
5. Temperature has a marked influence on the viscosity of a liquid. Increase in temperature results in rapid decrease in viscosity, about 2% per degree.

Procedure:

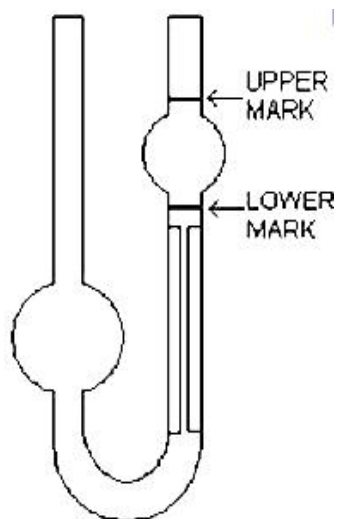
Rinse the given viscometer with acetone and dry it. Fix the viscometer vertically to a stand. Transfer 10 cm^3 of test liquid into a wider limb of the viscometer using burette. Record the time of flow in seconds. Repeat for 3 times.

Pour out the liquid, rinse the viscometer with acetone and dry it. Clamp it vertically to a stand and transfer the 10 cm^3 of water into the wider limb. As described above, record the time taken in seconds by the water to flow through the same distance. Determine the coefficient of viscosity of the liquid.

Report: The viscosity co-efficient of the given liquid = millipoise

OBSERVATIONS AND CALCULATIONS

FLOW TIME



Time flow in seconds	
Given Liquid	Fluid (Water)
1)	1)
2)	2)
3)	3)
Mean time (t _l) =	Mean time (t _w) =

Laboratory temperature: = °C

Density of water: (d_w) = g/cm³

Viscosity co-efficient of water (η_w) = millipoise

Density of given liquid: (d_l) = g/cm³

$$\text{Viscosity coefficient of liquid} = \frac{d_l \times t_l \times \eta_w}{d_w \times t_w} =$$

Answer the following viva questions:

1. What is Viscosity?
2. What is the SI Unit of viscosity coefficient?
3. How does the viscosity vary with temperature?
4. Write the expression used to calculate the viscosity coefficient of a liquid.
5. Why is the same volume of water and liquid used in the determination of viscosity coefficient of a liquid?
6. What is the name of the viscometer that is used?
7. Distinguish between Viscosity and Viscosity Coefficient.
8. How is viscosity related to density?

Colorimetric determination of Copper

Theoretical Discussion:

When a beam of light is passed through a medium, a portion of the radiant power (intensity), the beam gets absorbed. The extent of absorption depends upon the wavelength of the radiation and thickness (the concentration if solution used) of the medium. Not only has the wavelength at which the absorption takes place done the qualitative identification of the species, but also the amount of light absorbed permits the quantitative analysis.

Lambert Law: The law gives an important relationship between the amount of absorption of light and the thickness of the absorbing medium. It can be stated as *“each layer of equal thickness of a homogeneous absorbing medium absorbs an equal fraction of the radiant energy passing through it, i.e. rate of decrease of intensity of light (I) with the distance on passing through a medium is proportional to the initial intensity of the radiation”*. Thus

$$\frac{dI}{dx} = -kI \quad \text{or} \quad -\frac{dI}{I} = k \cdot dx$$

If the intensity of incident radiation is I_0 (when $x=0$), integration of the above expression gives: $I = I_0 e^{-kx}$

Beers Law: It states that *“Intensity of light decreases exponentially with the concentration of the light absorbing solute”*.

$$I = I_0 e^{-cx}$$

We sum up the two laws into a single equation as:

$$I = I_0 e^{-\epsilon c x}$$

Where c is the concentration (mol dm^{-3}) of the solute and ϵ is a constant characteristic to the solute but independent of its concentration is solution. This constant is known as *molecular absorption coefficient, molar absorbance index, molar extinction coefficient or more recently absorptivity*.

The above equation may be put in the form: $\log \left(\frac{I_0}{I} \right) = \epsilon c x$

The quantity $\log\left(\frac{I_o}{I}\right)$ is called *optical density*, D, or *absorbance*, A, or *extinction*, E. Thus

$$D \text{ (E or A)} = \log\left(\frac{I_o}{I}\right) = \epsilon c x$$

$$I = I_o e^{-D}$$

The ratio of the radiant power transmitted by a sample to the radiant power incident on the sample, i.e. the fraction of the light transmitted (I_o/I), is called *transmittance* of the solution. It is usually denoted by the letter **T**. Thus

$$\log\left(\frac{1}{T}\right) = \log\left(\frac{I_o}{I}\right) = \epsilon c x = A \text{ or } D$$

Percentage transmittance is given by 100T and percentage absorbance is 100(1 - T)

Principle:

A series of standard solutions of copper sulphate is treated with ammonia to get blue cupra-ammonium complex and is diluted to a definite volume. The absorbance of each of these solutions is measured at 620 nm, since the complex shows maximum absorbance at this wavelength.

The absorbance values are plotted against concentration to get a calibration curve. A known volume of the test solution is treated with strong ammonia and diluted to the same volume as above. The absorbance of this solution is measured at 620 nm and its concentration is determined from the calibration curve.



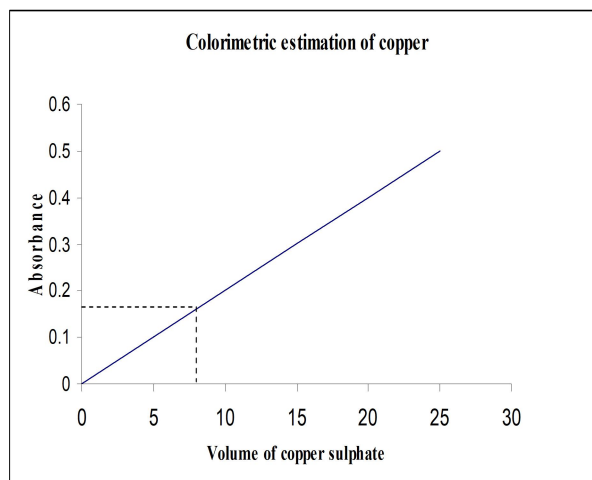
Procedure:

Transfer the given copper sulphate stock solution to a burette and draw out 5, 10, 15, 20 and 25 cm³ of solution into 50 cm³ volumetric flasks. Add 5 cm³ of ammonia solution to each and dilute up to the mark with ion exchange water. Record the absorbance of the above standard solutions and test solution with reference to the blank solution at 620 nm using photoelectric colorimeter.

Result: The wt. of Copper in the given test solution = g

OBSERVATION AND CALCULATION

Vol. of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	Absorbance
5	
10	
15	
20	
25	
Test solution	



Given:

The wt. of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ dissolved in 1000 cm^3 of water = g

249 g of $\text{CuSO}_4 = 63.54 \text{ g of Cu}$

Answer the following viva questions:

1. What is meant by transmittance?
2. What is absorbance?
3. What is the relation between absorbance and concentration?
4. What is blank solution?
5. What happens when ammonia is added to the copper sulphate solution in the colorimetric determination of copper?
6. Why is estimation of copper done at 620 nm?
7. What is path length?
8. State Lambert Beer's law or explain the principle involved in the colorimetric determination?

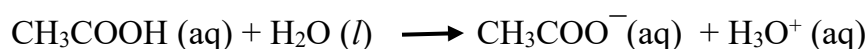
Determination of p^{ka} value of a weak acid using p^H meter

Introduction:

The strength of an acid is experimentally measured by determining its equilibrium constant or dissociation constant (K_a). Since strong acids are strong electrolytes, they are ionized almost completely in aqueous solution. It is not meaningful to study the ionic equilibrium of strong acids. Calculation of equilibrium and calculation of K_a is applicable only to weak acids and weak bases.

Principle:

Acetic acid ionizes feebly as



The dissociation constant or ionization, K_a is given by

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]}$$

p^{K_a} is modern method of expressing acid strengths; $p^{K_a} = -\log_{10} K_a$

p^{K_a} is determined by measuring the changes in pH of acid solution at different amounts of base added. During the titration of an acid with a base, the pH of the solution rises gradually at first, then more rapidly and until at the equivalence point, there will be very sharp increase in pH for a very small quantity of added base. The titration curve is obtained by plotting changes in pH at different amounts of the base added and the equivalence points are determined.

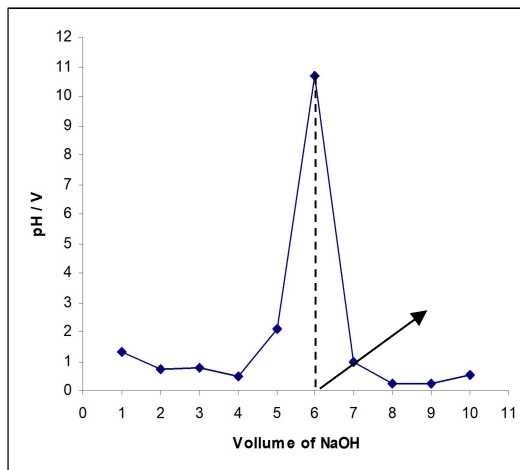
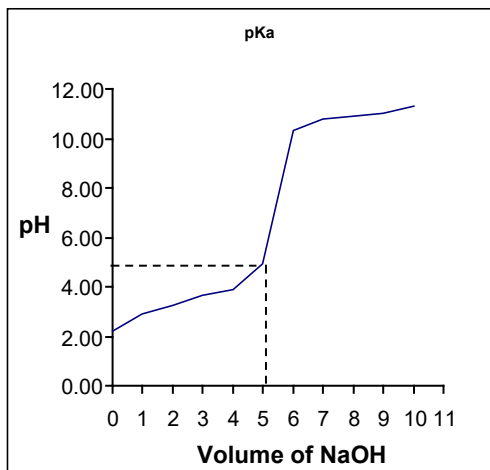
According to Henderson – Hasselbalch equation,

$$p^H = p^{ka} + \log_{10} \frac{[\text{salt}]}{[\text{acid}]}$$

At half equivalence point, $[\text{salt}] = [\text{acid}]$ and therefore, pH at half equivalence point gives the p^{K_a} of weak acid.

Procedure:

Pipette out 25 cm³ of the given weak acid into a 50 ml beaker, wash and place the glass and calomel electrode assembly into the acid solution and connect to pH meter. Record the initial pH of the acid. Fill the burette with sodium hydroxide. Add 0.5 cm³ sodium hydroxide from the burette to the acid solution and record the corresponding pH.



OBSERVATION AND CALCULATION

Volume of NaOH (cm³)	pH	ΔV	ΔpH	$\Delta pH / \Delta V$
0.0		—	—	—
0.5				
1.0				
1.5				
2.0				
2.5				
3.0				
3.5				
4.0				
4.5				
5.0				
5.5				
6.0				
6.5				
7.0				
7.5				
8.0				
8.5				
9.0				
9.5				
10.0				

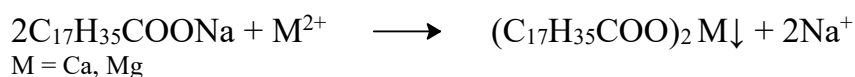
Report: The pKa of the given weak acid =

Answer the following viva questions:

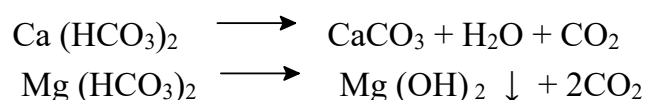
1. What is weak acid?
2. What is K_a ?
3. What is pK_a ?
4. How pH and pK_a is are related?
5. If the concentration of acid is changed, will the pK_a value of the acid be changed?
6. Explain the variation of pH during the determination of pK_a ?
7. What is Henderson equation for acid buffer?

Theoretical Discussion:

Hardness in water is due to the presence of bicarbonates, chlorides and sulphates of Ca and Mg. This term was applied to water which consumed a lot of soap without producing lather and was thus *hard* to wash in. The Ca^{2+} and Mg^{2+} ions in water form insoluble salts with soap, which is essentially sodium stearate.



Lather is not produced until the cations causing the precipitation of the sterates are completely removed and hence large quantity of soap to produce lather with hard water is needed. Hardness was formerly classified as ‘Temporary’ and ‘Permanent’. Temporary Hardness, due to the presence of bicarbonates of Ca and Mg can be largely removed by boiling, which converts the bicarbonates into the insoluble carbonates or hydroxides.

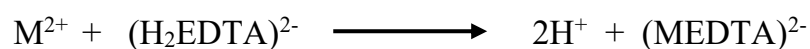


Permanent hardness is largely due to the presence of chlorides and sulphates of Ca and Mg cannot be removed by boiling. In order to relate the hardness to the chemical species in solution it is customary to refer the temporary hardness due to the carbonates and bicarbonates as carbonate hardness (CH) and the amount of hardness (permanent hardness) in excess of CH as non carbonate hardness (NCH). CH and NCH thus constitute the total hardness.

Principle:

The EDTA method is more accurate and is widely used for the estimation of hardness in water. The sample of water buffered to a pH of 9–10 is titrated against standard solution of disodium salt of EDTA using Eriochrome Black–T (Solochrome Black-T) as indicator.

Eriochrome Black-T is an azo dye capable of forming wine-red coloured complex with Ca^{2+} and Mg^{2+} at a pH of 9–10. As the EDTA is added, the free, uncomplexed Ca^{2+} and Mg^{2+} ions are complexed as metal ion–EDTA complexes.



M = Ca, Mg

These complexes are more stable than the indicator- metal ion complexes. Once the free metal ions are complexed, the $\text{Na}_2\text{-EDTA}$ forms complexes with Ca^{2+} and Mg^{2+} which are present as the respective indicator complexes. When all the hardness causing ions are complexed by $\text{Na}_2\text{-EDTA}$, the indicator is set free and since the indicator at this pH is blue in colour, the end point is the sharp change over in colour from wine-red to blue. The total hardness is first determined. The CH is removed by boiling and after the removal of the precipitate; the NCH in the filtrate is determined by complexometric titration with EDTA as before.

Total hardness – NCH = CH

Procedure:

Part A: Preparation of standard solution of $\text{Na}_2\text{-EDTA}$.

Take known weight of $\text{Na}_2\text{-EDTA}$ accurately and transfer to 100 cm^3 volumetric flask through funnel. Dissolve the crystals in ion exchange water, dilute up to the mark and mix well for uniform concentration.

Part B: Estimation of total hardness

Pipette out 25 cm^3 of the given water sample into a clean conical flask. Add 2 cm^3 of $\text{NH}_3\text{-NH}_4\text{Cl}$ buffer and add pinch of *Eriochrome black-T indicator*. Titrate against standard $\text{Na}_2\text{-EDTA solution}$ till the colour of the solution changes from wine red to clear blue. Repeat the titration for concordant readings.

Report: Total Hardness of the given water sample = ppm or mg/L

OBSERVATION AND CALCULATION

PART A: Preparation of standard solution of Di-sodium salt of EDTA

1. Weight of Bottle + disodium salt of Na₂-EDTA = g
2. Weight of Empty Bottle = g
3. Weight of disodium salt of Na₂-EDTA = g

$$\text{Molarity of Na}_2\text{-EDTA} = \frac{\text{Weight of Na}_2\text{-EDTA} \times 10}{\text{Molecular weight of Na}_2\text{-EDTA}} = \text{M}$$

Part B: Determination of Total Hardness of water sample

Burette Reading (cm ³)	I	II	III
Final Burette Reading			
Initial Burette Reading			
Volume of Na ₂ -EDTA consumed			

Volume of di-sodium salt of EDTA = cm³
(Concordant Burette Reading)

Conversion factor:

1000 cm³ of 1M di-sodium salt of EDTA $\equiv$ 100 g of CaCO₃

Answer the following viva questions:

1. What is normality?
2. What is molarity?
3. What is molarity?
4. What is hard water?
5. Write the reaction taking place with disodium salt EDTA with sample of water containing $\text{Ca}^{2+}/\text{Mg}^{2+}$ ions
6. Name the salts which cause carbonate hardness (temporary hardness) and non-carbonate hardness (permanent hardness).
7. Mention the difference between hard water and soft water.
8. Mention the harmful effects of hard water in industries.
9. What is permissible limit of hardness of potable water, and what happens if high degree of hard water is used for human consumption?
10. What is EDTA? Give its structure.

11. Name the indicator used in the estimation of total hardness of water. Give its structure, what is the endpoint?

12. How do you express the total hardness of water?

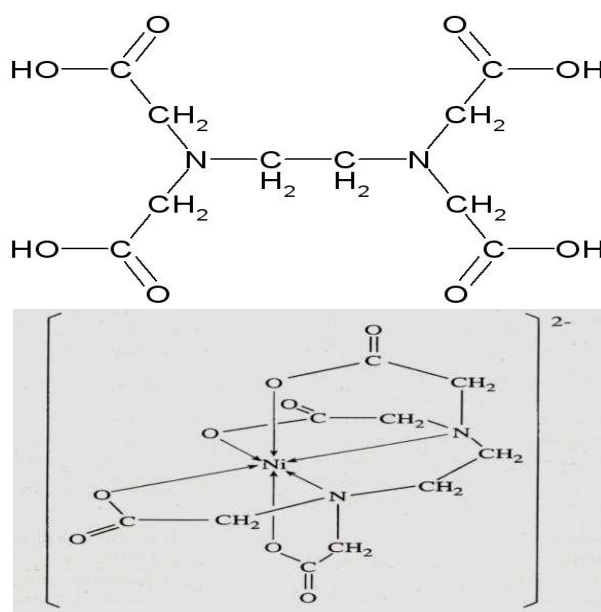
13. Why $\text{NH}_3\text{-NH}_4\text{Cl}$ buffer added in the determination of total hardness of water.

14. Why is disodium salt of EDTA preferred rather than EDTA to prepare Standard solution

Determination of Nickel Using EDTA by Complexometric Method.

Aim: To determine the percentage of Nickel in hydrated nickel (II) sulfate, $\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$ by EDTA titration.

Theory: EDTA is widely used to determine metals in complexometric (a volumetric analysis where the formation of a coloured complex is used to indicate the end point of a titration) titration as it forms stable complex with most metal ions. EDTA is a tetra Carboxylic Acid and in alkaline conditions, it exists as ions, which form 1:1 complex with metal ions like nickel (II) ions. A metal ion indicator (an organic dye which changes colour when it binds with metal ions) shows the endpoint of an EDTA complexometric titration. However, for a metal ion indicator to be suitable in EDTA titration, it must not bind as strongly with metal ions as EDTA does. Murexide is therefore suitable. The molecule ethylene diamine tetra acetic acid (EDTA) has the following structure.



It is often written as H_4Y where Y represent all the molecule except the four COOH hydrogen atoms. In alkaline conditions the molecule forms the Y^{4-} ion. This ion reacts in a one to one stoichiometric ratio with a range of divalent metal ions (Mg^{2+} , Ni^{2+} etc). The EDTA ion “wraps around” the metal ion forming a metal - EDTA octahedral complex which is represented by the structures shown.

Since EDTA forms stable complexes with most metal ions, it is widely used to determine metals in what are known as complexometric titrations. The reaction of nickel (II) ions with EDTA can be represented as



The end-point of an EDTA complexometric titration can be detected by means of a metal ion indicator – an organic dye which changes colour when it binds with metal ions. For it to be suitable in an EDTA titration, the indicator must bind less strongly with metal ions than does EDTA. Murexide is one such indicator.

Reagents required:

Disodium salt of EDTA solution (0.01 M): 0.9305 g of AR disodium dihydrogen ethylene diamine tetra acetate in deionized water diluted to 250ml. Murexide Indicator (0.6%): 0.4 g of the dye in 100ml ethanol.

Procedure:

Part - A Preparation of std. solution of Na₂-EDTA

Take known weight of di-sodium salt of EDTA accurately and transfer to 100 cm³ volumetric flask through funnel. Dissolve the crystals in ion exchange water, dilute up to the mark and mix well for uniform concentration.

Part - B Determination of Nickel in the sample solution:

Pipette out 25 cm³ of the given Nickel sulphate sample solution into a clean conical flask. Add 2cm³ of NH₃-NH₄Cl buffer and a pinch of Murexide Indicator. Titrate against standard Na₂-EDTA solution taken in the burette till the color of the solution changes from orange to violet. Repeat the titration till concordant values are obtained. Let the volume of Na₂-EDTA consumed be Y cm³

Note: During the titration of a solution containing Ni²⁺ ion with Na₂-EDTA in the presence of Murexide Indicator, disodium salt of EDTA reacts first with Ni²⁺ion, and finally with Ni- Murexide complex. Since the Ni- Murexide complex is blue in color and the free indicator is blue between pH 7 and 11, the color of the solution changes from wine red to Violet at the end point.

\Result: Percentage of Nickel (II) is present in the sample of**Part**

– A: Preparation of standard solution of Na₂ [EDTA]

Weight of Na₂ EDTA salt =g

$$\text{Molarity of Na}_2\text{ EDTA} = \frac{\text{Weight of EDTA salt} \times 1000}{\text{Mol.wt of EDTA salt} \times 250}$$

$$= \dots\dots\dots \times 4 \\ 372.24$$

$$= \dots\dots\dots \text{M} = \text{X}$$

PART B: Estimation of Percentage of Nickel (II)

Burette: Standard disodium salt of Na₂ [EDTA] solution

Conical flask: 25ml of Nickel (II) sulphate solution + 2 ml of Ammonia- ammonium chloride

Buffer solution (NH₃-NH₄Cl) {pH = 10}

Indicator: Pinch or 2-3 drops of Murexide indicator (C₈H₈N₆O₆)

Endpoint: Orange to Violet

Burette Level	I	II	III	Mean V in ml
Final Burette reading				
Initial Burette reading				
Difference in ml				

Calculation:

Weight of Nickel Sulphate in 250cm³ = g (W_{NiSO₄})

1000 cm³ of 1M Na₂ [EDTA] = 58.69g of Ni (M.W of Ni is 58.69)

y cm³ of M Na₂ [EDTA]= 58.69 × X × Y / 1000 g of Ni = A g of CaO

25cm³ of Nickel Sulphate Solution contains 'A' g of Ni

250 cm³ of Nickel Sulphate Solution contains A × 10 g of Ni = 'Z' g of Ni

If W is weight of Nickel Sulphate taken,

Then percentage of Ni in the sample of = Z × 100/W_{Nickel sulphate} (g)

EXPERIMENT: 3

DATE:

**Determination of Percentage of Copper in Brass Using Standard
Sodium Thiosulphate Solution (Iodometric Method)**

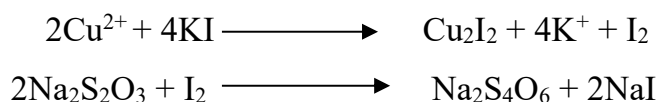
Principle:

The chief constituents of brass alloys are copper and zinc. It also contains small quantities of tin, lead and iron. The percentage composition of a typical brass alloy is

Cu: 50 – 90%, Zn : 20 – 40 % Sn: 0 – 6 %, Pb: 0 – 2 %, Fe: 0 – 1 %

A solution of brass is made by dissolving the sample in minimum amount of nitric acid. Excess of nitric acid is destroyed by boiling with urea. The solution is neutralized and made faintly acidic. Potassium iodide solution is

added when the cupric ions oxidize iodide to iodine. The iodine liberated is titrated against sodium thiosulphate using starch as indicator. The volume of sodium thiosulphate consumed is a measure of the amount of the copper present.



Procedure:

Part A: Dissolution of the brass sample

Take a known weight of the given brass sample accurately and transfer it into a clean beaker. Add a test tube of 1:1 Nitric acid. Provide the glass rod and **boil till the alloy dissolves completely**. Add 1 test tube of ion exchange water and gently boil till all brown colored oxides of nitrogen are expelled out completely. Add a spatula of urea and continue boiling for some more time to expel all the fumes. Transfer the blue coloured solution quantitatively into 250 cm³ standard flask with a glass funnel. *Wash the beaker 3 to 4 times with a little quantity of ion-exchange water. Dilute the solution up to the mark, and mix well for uniform concentration.*

Part B: Estimation of copper

Pipette out 25 cm³ of the above solution into a clean 100 cm³ conical flask. Add dilute NH₄OH drop wise until a slight bluish white precipitate is formed. Dissolve the precipitate by adding about ½ test tube of dilute acetic acid. Add 1 test tube of 10% KI solution and one test tube of ion ex-change water, shake and titrate the iodine liberated against standard sodium thiosulphate solution till the mixture turns light yellow. Add 2 cm³ of starch indicator and continue the titration till the blue colour just disappears. Repeat the titration for concordant readings.

Report: Percentage of copper in the given Brass sample = _____ g

OBSERVATION AND CALCULATION

PART A: Preparation of Brass solution

Weight of Brass = g

Part B: Estimation of copper

Burette Reading (cm ³)	I	II	III
Final Burette Reading			
Initial Burette Reading			
Volume of Na ₂ S ₂ O ₃ consumed			

Volume of Na₂S₂O₃ = cm³

Normality of Na₂S₂O₃ = N

Conversion factor:

1000 cm³ of 1N Na₂S₂O₃ $\equiv$ 63.54 g of Cu

Answer the following viva questions

1. What is **Brass**? What are the constituents of brass?
2. What is the principle of **Iodometric** titration?
3. How brass is solution prepared?
4. What is the purpose of adding urea to the brass solution?
5. Why is Ammonium Hydroxide added to brass solution?
6. Why **KI** is added to the brass solution?
7. Why the blue solution of brass turns brown by the addition of KI?
8. Why is starch indicator added towards the end point in Iodometric titrations?
9. What is the white precipitate left at the end point in the determination of copper in brass?
10. Write the reaction-taking place with Cu^{++} ions and Potassium iodide.
11. Write the reaction-taking place with sodium thiosulphate and Iodine.
12. What is the equivalent weight of sodium thiosulphate?

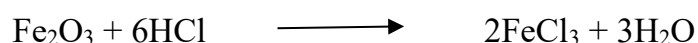
**Estimation of Percentage of Iron in the given Rust Solution Using Standard
K₂Cr₂O₇ Solution (external indicator method)**

Theoretical Discussion:

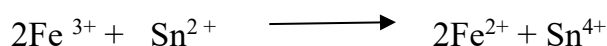
Iron often occurs in the ferric state, as a mixture of ferrous and ferric iron in ores and other materials. For the volumetric determination of iron, the ferric iron must be first quantitatively reduced to the ferrous state using some reducing agents. Many iron ores are brought into solution with concentrated hydrochloric acid. In such cases the reducing agent frequently employed is stannous chloride.

Principle:

Hematite is an important ore of iron containing mainly Ferric oxide, Fe₂O₃ and small amounts of silica (SiO₂). The given Hematite sample is treated with concentrated hydrochloric acid, the insoluble silica is filtered off and the filtrate is used for the estimation of iron.



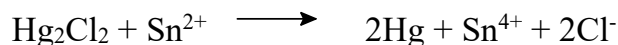
The hot iron solution (70–90°C) is reduced by adding concentrated stannous chloride solution drop wise from a burette, with stirring.



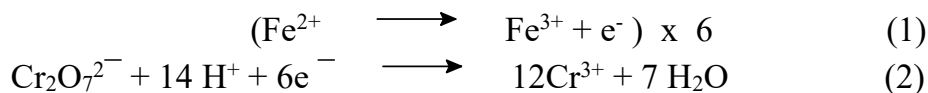
The solution is then rapidly cooled under the tap to about 20°C with protection from the air and the slight excess of stannous chloride present is removed by adding saturated solution of mercuric chloride rapidly in one portion and with thorough mixing. A slight silky white precipitate of mercurous chloride should be obtained:



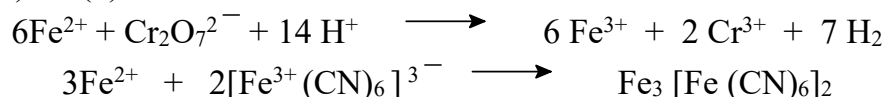
The oxidizing agent has no appreciable effect upon the small amount of mercurous chloride in suspension. If a heavy precipitate forms, or a grey or black precipitate is obtained, too much stannous chloride has been used: the results are inaccurate and the reduction must be repeated. Finely divided mercury reduces dichromate ion and also slowly reduces ferric ions in the presence of chloride ions



After the addition of the mercuric chloride solution the whole is allowed to stand for five minutes, and the mixture is titrated slowly and constant stirring with standard potassium $\text{K}_2\text{Cr}_2\text{O}_7$ solution using potassium ferricyanide as an external indicator.



Adding (1) & (2)



Procedure:

Part A: Preparation of Standard Potassium dichromate solution.

Take known weight of potassium dichromate crystals accurately and transfer to 100 cm³ volumetric flask. Dissolve in a minimum quantity of ion exchange water and dilute up to mark and mix well.

Part B: Estimation of Iron

Pipette out 25 cm³ of the given hematite ore solution into a clean conical flask. Add ½ test tube of 1: 1 HCl and heat the solution nearly to boiling. Add **stannous chloride** to the hot solution drop wise till the yellow solution turns colourless. Add 2 more drops of stannous chloride to ensure complete reduction. Cool and add ½ test tube of saturated mercuric chloride solution rapidly. A silky white precipitate of **mercurous chloride** is formed. Add one test tube of ion exchange water. Titrate the contents of the flask against standard potassium dichromate solution using **potassium ferricyanide** as an external indicator. A drop of the reaction mixture is withdrawn using a clean glass rod and brought in contact with the indicator drop. The end point is reached when the indicator fails to give blue coloration. Repeat the titration for concordant readings.

Report: Percentage of Iron in the given Hematite ore solution =

OBSERVATION AND CALCULATION

PART A: Preparation of standard solution of Potassium dichromate

1. Weight of Bottle + $K_2Cr_2O_7$ = g
2. Weight of Bottle = g
3. Weight of $K_2Cr_2O_7$ = g

$$\text{Normality of } K_2Cr_2O_7 = \frac{\text{Weight of } K_2Cr_2O_7 \times 10}{\text{Eq. weight of } K_2Cr_2O_7} = \text{N}$$

Part B: Estimation of Iron:

Burette Reading (cm ³)	I	II	III
Final Burette Reading			
Initial Burette Reading			
Volume of $K_2Cr_2O_7$ consumed			

Volume of $K_2Cr_2O_7$ = cm³

Conversion factor: 1000 cm³ of 1N $K_2Cr_2O_7$ $\equiv$ 55.85 g of Fe

Answer the following viva questions

1. What is Hematite? Give its composition.
2. Why is hydrochloride acid added to ferric chloride solution during reduction?
3. What is the role of stannous chloride in the estimation of iron?
4. Explain the role of mercuric chloride.
5. Why is there a little excess of stannous chloride added? What happens if excess of stannous chloride is not removed?
6. Why potassium ferricyanide is cannot be used as an internal indicator in the analysis of hematite ore?
7. Name the blue complex formed between Fe^{2+} and potassium ferricyanide Indicator.
8. Why the iron solution cooled immediately after the reduction is completed with SnCl_2 ?
9. Write the reaction-taking place with stannous chloride and ferric chloride.
10. Write the reaction-taking place with stannous chloride and mercuric chloride.

Determination of Chemical Oxygen Demand (COD) of the given Industrial waste water sample

Theory: The chemical oxygen demand (COD) is a mixture of the oxygen equivalent to that portion of organic matter present in the waste water sample that is susceptible to oxidation by potassium dichromate. This is an important and quickly measured parameter for stream sewage and industrial waste samples to determine their pollution strength.

According to the American society of testing and material (ASTM) *“COD is defined as the amount of oxygen (expressed in mg/l) consumed under specified conditions in the oxidation of organic and oxidisable inorganic matter, corrected for the influence of chlorides”*.

(Since in the COD test, both the biologically oxidisable and the biologically inert matter are oxidized, the COD value for sample is always higher than BOD value.)

Principle:

The principle involved in the determination of COD is that when the waste water sample is refluxed with a known excess of potassium dichromate in a 50% H_2SO_4 solution in presence of AgSO_4 (as catalyst) and HgSO_4 (to eliminate interference due to chloride), the organic matter of the sample is oxidized to water, CO_2 and ammonia. The excess of dichromate remaining unreacted in the solution is titrated with a standard solution of Ferrous Ammonium Sulphate using Ferroin as indicator.

Procedure:

Part A: Preparation of standard Mohr's salt solution (FAS solution)

Take known weight of Mohr's salt accurately into 100 cm^3 volumetric flask. Add $\frac{1}{2}$ test tube of dilute sulphuric acid and dissolve the crystals. Dilute the solution with ion exchange water up to the mark and shake well for uniform concentration.

Part B: Estimation of COD

Back titration:

Pipette out 25 cm³ of the water sample into a conical flask. Add 25 cm³ of standard potassium dichromate solution followed by 1½ test tube of dilute sulphuric acid containing silver sulphate with constant shaking. Add 2–3 drops of ferroin indicator and titrate against standard Mohr's salt solution until the solution turns from blue green to reddish brown. Repeat the titration for concordant readings.

Blank titration:

Pipette out 25 cm³ of standard potassium dichromate solution. Add 1½ test tube of dilute sulphuric acid containing silver sulphate followed by 2–3 drops of ferroin indicator. Titrate against standard Mohr's salt solution until the colour turns from blue green to reddish brown.

Report: COD of given waste water = ppm

OBSERVATION AND CALCULATION

PART A: Preparation of standard FAS solution

- | | | |
|---------------------------|---|---|
| 1. Weight of Bottle + FAS | = | g |
| 2. Weight of Empty Bottle | = | g |
| 3. Weight of FAS | = | g |

$$\text{Normality of FAS} = \frac{\text{Weight of FAS} \times 10}{\text{Eq. weight of FAS}} = \text{N}$$

Part B: Estimation of COD:

Back Titration

Burette Reading (cm ³)	I	II	III
Final Reading			
Initial Reading			
Volume of FAS			

Blank Titration

Burette Reading (cm ³)	I
Final Reading	
Initial Reading	
Volume of FAS	

$$\text{Volume of FAS} = \text{cm}^3 (v_1)$$

$$\text{Volume of FAS} = \text{cm}^3 (v_2)$$

$$\text{Volume of FAS } (v_2 - v_1) = \text{cm}^3$$

Conversion factor: 1000 cm³ of 1N FAS = 8 g of Oxygen

Answer the following questions

1. What is Chemical Oxygen Demand?
2. What is the role of silver sulphate in the determination of COD?
3. What is the role of mercuric sulphate in the determination of COD?
4. Name the indicator used in COD experiment. Write the structure, IUPAC name and colour change at the end point of indicator.
5. Why is Ferrous ammonium sulphate [Mohr's salt] prepared in dil. H_2SO_4 ?
6. Calculate the equivalent weight of FAS.
7. What is the equivalent weight of potassium dichromate?
8. Why is potassium dichromate used as a primary standard?

DEMONSTRATION EXPERIMENTS

EXPERIMENT: 1

DATE:

Flame Photometric estimation of Sodium in the given sample of water

Flame photometry:

Introduction:

Flame photometry is based the measurement of intensity of the light emitted when a metal is introduced into a flame. The wavelength of the colour tells us what the element, and the colours intensity tells us how much of the element is present.

Flame photometry is also named as flame emission spectroscopy because of the use of a flame to provide the energy of excitation to atoms introduced into the flame.

Flame photometry is used for the determination of elements such as Na, K, Li, Ca, Mg, Sr, and Ba. The measurement of these elements is very useful in medicine, agriculture and plant science. Flame photometry is also successful in determining certain transition elements, such as Cu, Fe and Mn. Flame photometry is a simple, rapid method for the routine determination of elements that can be easily excited.

Principle:

When a liquid sample containing a metallic salt solution is introduced into a flame the processes involved in flame photometry are complex, but the following is a simplified version of the events.

- i) The solvent is vaporized, leaving particles of the solid salt.
- ii) The salt is vaporized or converted into the gaseous state.
- iii) A part or all the gaseous molecules are progressively dissociated to give free neutral atoms or radicals. These neutral atoms are excited by the thermal energy of the flame. The excited atoms, which are unstable, quickly emit photons and return to lower energy state, eventually reaching the unexcited state. The measurement of the emitted photons i.e., radiation, forms the basis of flame photometry. When flame photometry is employed as an analytical tool, the wavelength of the radiation coming from a flame tells us about the elements which are

present in that flame. Also, the intensity of the radiation enables us to know the amounts of those elements present.

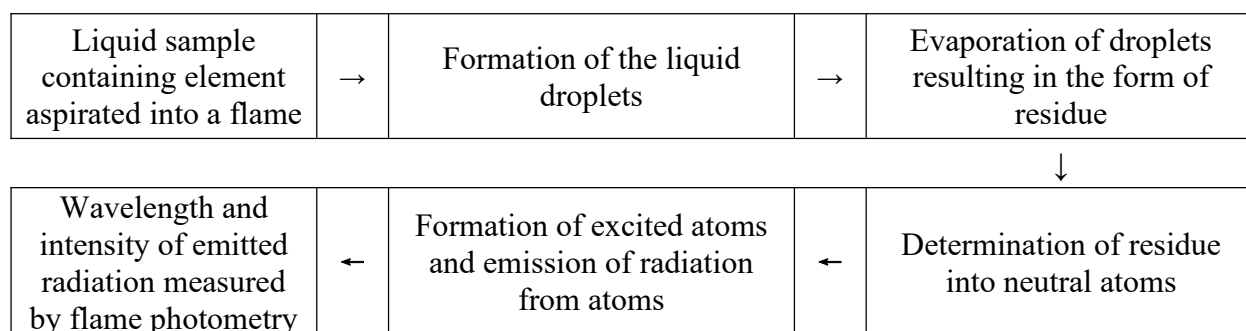
The fraction of free atoms that are thermally excited is governed by a Boltzmann distribution which is as follows.

$$\frac{N^*}{N_0} = A e^{-\frac{\Delta E}{kT}}$$

Where N^* is the number of excited atoms

N_0 is the total number of atoms remaining in the ground state.

A is a constant for a particular element, ΔE , the difference energies of the two levels, k the Boltzmann's constant, and T the temperature of flame.



OBSERVATION AND CALCULATION:

Volume of Na taken in cm ³	Concentration in ppm	Flame photometric reading
2		
4		
6		
8		
10		
Unknown		

Volume of Na ion solution	Flame photo meter reading
1	5
2	10
3	15
4	20
5	25

The weight of NaCl dissolved in 1000 cm³ of water = g

Report : The wt. of sodium in the given test solution = g

Answer the following viva questions

1. What is meant by Flame photometry?
2. What is the basic principle which supports for flame photometry?
3. Mention the different types of flames?
4. Mention the colour of the flame? Writ the following elements
 - a. Sodium, b. calcium, c. Potassium, d. Barium
5. What is the function of filter?
6. What is the wave length of sodium filter?

EXPERIMENT:2

DATE:

Synthesis of nonmaterial by precipitation method
